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CHAPTER

Land Resources

Chapter Outline

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INTRODUCTION

Land is a major constituent of lithosphere and is one of the main components of biosphere, besides air and water. The biosphere forms a natural environment for plants and animals including man. Land forms about 29.1 per cent of the earth surface covering an area of about 148,950,800 sq km. It is the most precious resource because it is put to diverse uses by man and is the source of many materials essential to man and other organisms. Performance of various natural activities need space for their location and development which is provided by land. Many life sustaining biogeochemical cycles take place in the land. Land supports large forest areas and many water bodies, such as rivers, lakes, ponds *etc.* Various purposes, for which man uses land include agriculture and horticulture for food production; energy production; extraction of minerals, oil, coal, natural gas; human dwellings; commercial and industrial purposes; waste disposal *etc.* India with a land area of about 2.4 per cent of the global land area, supports about 15 per cent of world's population.

PATTERN OF LAND USE

The pattern of land use varies from country to country. The pattern of land use distribution in the world is 30 per cent forest land, 26 per cent pasture land, 11 per cent cropland and the rest 33 per cent as land like tundra, desert,

bare rock, snow *etc*. In India, the so called agricultural country, more than two-fifth of the land is agricultural land. The pattern of land distribution in India is 43.6 per cent agricultural land and cultivated land, 14.6 per cent permanent pasture and meadows, 12 per cent culturable wasteland, 11.5 per cent forest, 8.0 per cent barren and unculturable land, 5.3 per cent urban land and 5 per cent for which no proper information is available (Fig. 8.1).

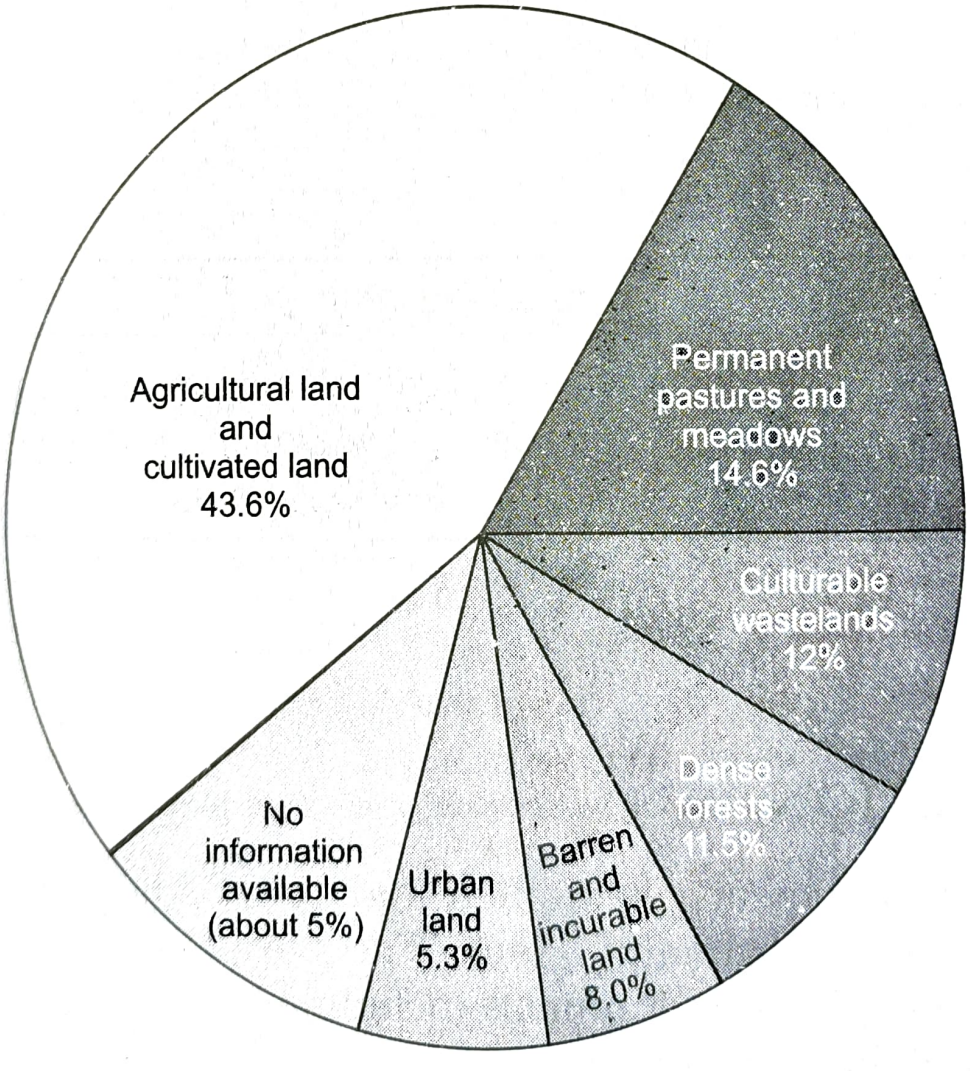


Fig. 8.1. Pattern of land distribution in India (approximates)

LAND AS A RESOURCE

Land is a storehouse of valuable resources upon which man depends for food, shelter, clothing, movement, security and many other of his needs and greed. The surface layer of the land is called soil. About four-fifth of the land area is covered by the soil. The word soil has been derived from the Latin word *solum* – means upper crust of the earth. The soil in general is defined as the upper layer of the earth which is differentiated into various horizons and is capable of supporting plant life. The top horizon—the fertile region of the soil is classified as a renewable resource because it keeps on continuously regenerating, changing or developing by natural processes at a very slow rate. The soil is formed by wear and tear of the earth's crust, a process which has been going on through the ages. Many biogeochemical cycles operate in the soil due to presence of a variety of microorganisms including many forms of bacteria, fungi, plants and animals. The upper

limit of the soil is water or air and the lowest limit is difficult to define but is normally thought of as the lower limit of common rooting depth of native perennial plants, a boundary that is shallow in deserts and tundra and deep in humid tropics.

Soil is a dynamic layer of earth's crust. It takes about 200–1000 years to form 2.5 cm of soil, depending upon various factors, such as climate, soil type, processes involved in decomposition of parental rocks *etc.* The soil productivity is determined by many soil factors, such as soil texture; availability of water; permeability; soil water porosity; soil pH; organic matter and inorganic nutrient contents; cation exchange capacity; microbial population *etc.* The topography and biotic factors also play an important role in determining the soil conditions. Soil plays a key role in determining the quality and composition of the biosphere. Infact, biosphere develops over the soil and soil provides nutrition to the plants and abodes the microorganisms. The land is vital to our existence due to the following facts:

1. It preserves terrestrial biodiversity and genetic pool.
2. It regulates water and carbon cycles.
3. It acts a store house of basic resources like ground water, minerals and fossil fuels.
4. It becomes a dump of solid and liquid wastes.
5. It forms a basis for human settlement and transport activities.
6. Its top soil horizon, which is only a few cm thick, supports all plant growth and is hence the life support system for all organisms, including man.

LAND DEGRADATION AND ITS CAUSES

Land degradation refers to deterioration or loss of fertility or productive capacity of the soil. All modern growth oriented activities are directly or indirectly causing impacts on land. According to UN studies, 23 per cent of all usable land, excluding mountains, deserts, polar regions *etc.* has been degraded to such an extent that its productivity has been affected. The main causes of this are deforestation, water logging, excessive use of chemicals, agricultural mismanagement, soil salinity, fuelwood consumption, overgrazing, planting unsuitable crops, poor crop rotation, poor soil and water management, frequent use of heavy machinery, such as tractors, heavy industrialisation and urbanisation.

Rapid rate of increase of human population coupled with man's increased demand of food and other needs, has brought the land under pressure. The per capita land resource is showing marked decrease in many thickly populated countries. In India, it may go down from 0.33 hectare to 0.25 hectare in near future. Equally responsible is heavy industrialisation, demanding space and raw materials and causing

pollution. In Asian tropics, land degradation is a major problem. It is estimated that about one-third of the irrigated agricultural land, about half of rainfed crop land and about three-fourth of the pastoral land are degraded. In India alone about 170 million hectares of productive land is degraded to a greater or lesser extent. The cause being decimation of much of the forests in the lowland, forest conversion in uplands for timber to support the industries else where, mining, energy generation through large hydro-electric projects *etc*. Ecological problems and land degradation have accelerated rapid marginalisation of a large section of the rural communities and needs to be viewed upon seriously.

In India, land degradation is a key issue, especially in rural environment. Some of the important issues include improving fertility of degraded infertile soils, improving the sub-soil water balance depleted due to over-exploitation of ground water for agriculture, saline and alkali soil (usar land) formation in areas under green revolution agriculture especially in Haryana, Punjab, Western Uttar Pradesh *etc*. The changing land use patterns, such as utilisation of agricultural land for urbanisation and growing soil degrading plants for industrial requirements *etc*. also need to be paid great attention. Soil erosion and contamination of soil with industrial wastes like fly ash, press mud or heavy metals, all causing soil degradation, also need to be looked upon.

many tourists to be stranded in these places or on the way causing many problems and require measures for relief. The landslides can be checked by the construction of drainage ditches and walls along the hill slopes.

SOIL EROSION

Soil erosion refers to the loss or removal of the top fertile layer of the soil by action of wind, water, ocean waves, glaciers and human activities. It results in the loss of soil fertility. Causes of soil erosion are many but deforestation without reforestation, overgrazing by cattle, surface mining without land reclamation, irrigation techniques leading to salt build-up, waterlogging of soil, farming on land with unsuitable terrain, soil compaction by crampling of cattle and heavy machinery, such as tractors, threshers *etc.*, cause the top soil vulnerable to soil erosion. Moreover, loss of green cover, strong winds and chemical pollution also severely affect the fertility of soils and the soil's ability to act as buffer and filter for pollutants, regulator for water and nitrogen cycle and habitats for biodiversity. Other factors affecting soil erosion are cultivation on hill slopes; lack of proper irrigation; duration, intensity and distribution of rainfall and floods; social factors; topography; drainage area; kind of vegetation cover type and extent of covering; presence or absence of well defined channels; hydraulic properties and permeability *etc.* of subsoil. It is estimated that about one-third of world's cropland is getting eroded. World's two-third of the seriously eroded and degraded land lies in South Africa.

The soil erosion is basically of two types:

1. **Geological Erosion or Natural Erosion:** Geological erosion is caused by gradual removal of the top soil by natural processes which bring about equilibrium between physical, hydrological and biological activities and maintain a natural balance between erosion and renewal.
2. **Accelerated Erosion:** Accelerated erosion is caused due to man-made activities. The rate of accelerated erosion is much faster than the rate of soil formation. It is caused chiefly by population pressure, deforestation, change in land use pattern, mismanaged agricultural practices, overgrazing, intensive cropping, mining and constructions leading to development *etc.* The soil erosion accounted by overgrazing is 35 per cent and that of the deforestation is 30 per cent. Unsustainable methods of farming cause 28 per cent of the soil to become eroded.

Soil erosion is very extensive in Australia, Central Africa, USA, Spain, Russia, Nepal, India *etc.* In India, extreme soil erosion by wind and water, is resulting 48,000 hectare of land to become eroded every year. Soil

erosion has resulted in greatest menace to agriculture. Water erosion is common in Maharashtra, Karnataka, Gujarat, Andhra Pradesh, Punjab, Uttar Pradesh, Orissa and Bihar. Wind erosion is common in Rajasthan, Punjab, Madhya Pradesh, Uttar Pradesh *etc*. Huge sandy areas in Punjab and Haryana are the example of wind erosion. The 700 year old Thar of Rajasthan, which was once a sea coast rich in vegetation and civilisation, has been converted into India's largest desert. In India overgrazing alone is responsible for 18.5 per cent of global loss of soil fertility. It is estimated that the fertile top soil of Shivalic hills, which took about 24,000 years for its ecological construction of fertile top soil, is being lost by flooding at the rate of 6 cm per year. In North East Hill States of India, shifting cultivation, *i.e.*, slash and burn or jhum cultivation is the main cause of soil erosion.

PREVENTION OF SOIL EROSION

Soil erosion can be prevented by various methods of soil conservation. The best course is large-scale tree plantation of soil binding shrubs and development of grasslands. Plant roots hold the soil and prevent erosion caused by flood, wind and water. Leaves of the plants, especially of trees, intercept the rain and break the impact of rainfall water. Vegetation, as a whole, protects the soil surface from action of wind, run off water and solarisation and reduces run off of organic matter. Planting trees, shrubs or grasses stabilises the soil and checks soil erosion. Grasses, in particular, can be grown on soils which are otherwise not suitable for the cultivation of crops. Grasses are also useful for protecting bunds, water ways, highway alignments, badly eroded areas, gully control and for stabilising sand dunes. Farm forestry is another important effort in soil conservation. Planting of trees as wind breaks in the shelter belts is necessary to check wind erosion. Erosion caused by floods can be checked by small dams. Another method to check soil erosion caused by water is to develop a well controlled drainage system which could prevent the uncontrolled water flow. Application of various agronomic methods, forestry management measures or change in land use patterns and mechanical or engineering methods, to check accelerated soil erosion, have also been applied.

Erosion by the sea waves and currents has been checked to a large extent in Orissa, Andhra Pradesh, Kerala and some other coastal areas by the construction of broad walls of stone along the coasts with heavy plantation of *Casuarina* trees. In hill slopes or terraced fields, the alternation of crop beds with erosion resistent vegetation is practiced so that the soil gets stabilised. In cyclone-prone and desert areas, the effect of strong winds is checked by putting barriers of shrubs and trees.